

COLLABORATIVE DISCUSSION 1:
INDUSTRY 4.0 AND 5.0 IN AUTOMOTIVE -
BMW AND JAGUAR LAND ROVER

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Unit 3

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INITIAL POST

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BMW's AI assisted maintenance system at one of its key plants in Germany monitors the production line and enables early detection of faults, which translates into significant disruption avoidance in their vehicle production (BMW Group, 2023). In 2025 they also reported that their AI quality-control system analyses real-time production data to recommend custom inspections to the vehicle assembly line (BMW Group, 2025). These are examples from an industry I am deeply interested in, where Industry 4.0 delivers smart factories which can improve quality, speed and reliability of the production lines.

However, that very same efficiency requires highly reliable and disruption-resilient information systems. In 2025, Jaguar Land Rover suffered a cyber incident that severely disrupted retail and production activities (Jaguar Land Rover, 2025). Young (2025) reported that the shutdown lasted weeks, affected suppliers and staff, and required government attention. Reuters (2025) later reported cyberattack-related costs of £196 million. These are a few of many examples that are documented; the more connected and smart automotive manufacturing is, information systems become more central and can quickly become a production, customer, supplier and reputational crisis when failures arise.

This links to Metcalf's (2024) argument that Industry 5.0 responds to technology centered Industry 4.0 by incorporating resilience and human-centricity. Then, there is the ethics side where AI and robotics may raise productivity, but also widen the digital divide and reduce human labour in factories (Korinek and Stiglitz, 2021; OECD, 2023).

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